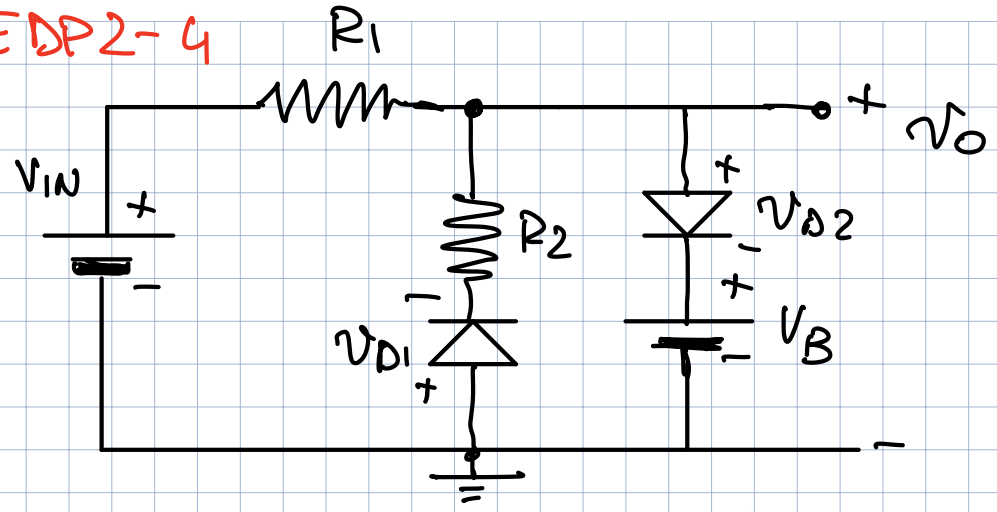
EDP2-4



$R_1 = 1K\Omega$

$R_2 = 1K\Omega$

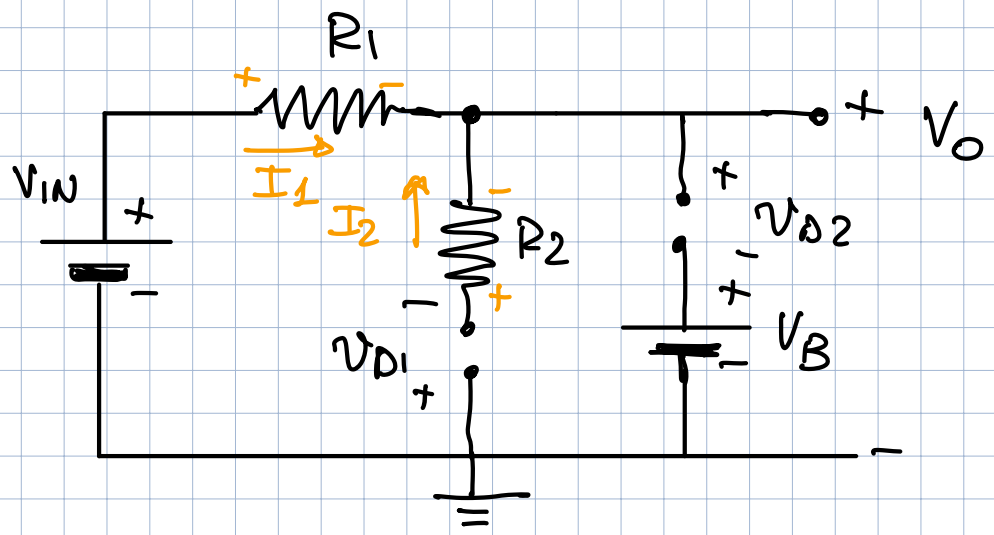
$V_B = 15V$

$V_{ON} = 1V$

TROVARE v_o QUANDO $V_{IN} = -20V$, $v_{IN} = 10V$, $v_{IN} = 30V$

FACOLTATIVO! TROVARE $v_o = f(V_i)$

HP D1 = OFF e D2 = OFF



$I_1 = I_2 = 0$!

$V_o = V_{IN}$

$v_{D1} = -V_{IN}$

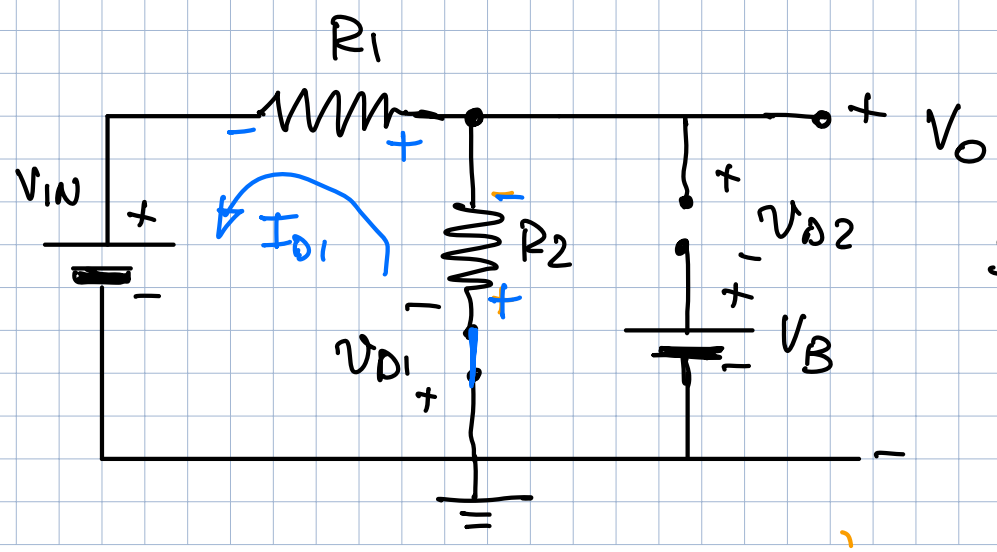
$v_{D2} = V_o - V_B$

$= V_{IN} - V_B$

a) $V_{IN} = -20V$

$\Rightarrow v_{D1} = -V_{IN} = 20V \Rightarrow D1 = ON$

$v_{D2} = V_{IN} - V_B = -35V \Rightarrow D2 = OFF$



$I_{D1}(R_1 + R_2) + V_{IN} = 0$

$I_{D1} = -\frac{V_{IN}}{R_1 + R_2} = \frac{20V}{2K\Omega} = 10mA$

> 0 OK "ON"

$V_o = -R_2 I_{D1} = -1K\Omega \cdot 10mA = -10V$

$\Rightarrow v_{D2} = V_o - V_B = -25V < 0$ OK $D2 = OFF$

QUINDI $V_{IN} = -20V$
$D1 = ON$
$D2 = OFF$
$V_o = -10V$

b) $V_{IN} = 10V$ (DAL CASO $D1 = D2 = OFF$)

$\Rightarrow v_{D1} = -V_{IN} = -10V \Rightarrow D1 = OFF$ ✓

$v_{D2} = V_{IN} - V_B = -5V \Rightarrow D2 = OFF$ ✓

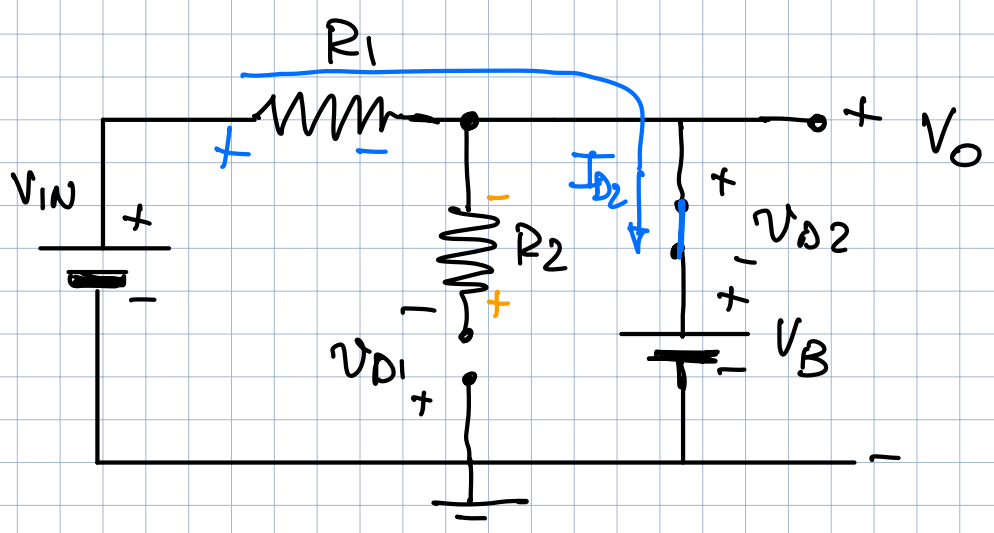
QUINDI $V_{IN} = 10V$
$D1 = OFF$
$D2 = OFF$
$V_o = V_{IN} = 10V$

c) $V_{IN} = 30V$ (DAL CASO $D1 = D2 = OFF$)

$\Rightarrow v_{D1} = -V_{IN} = -30V \Rightarrow D1 = OFF$ ✓

$v_{D2} = V_{IN} - V_B = 15V \Rightarrow D2 = ON$

$\Rightarrow$ HP $D1 = OFF$, $D2 = ON$



$I_{D2} = \frac{V_{IN} - V_B}{R_1} = \frac{30V - 15V}{1K\Omega} = 15\,mA > 0$

OK $D2 = ON$

$V_o = V_B$

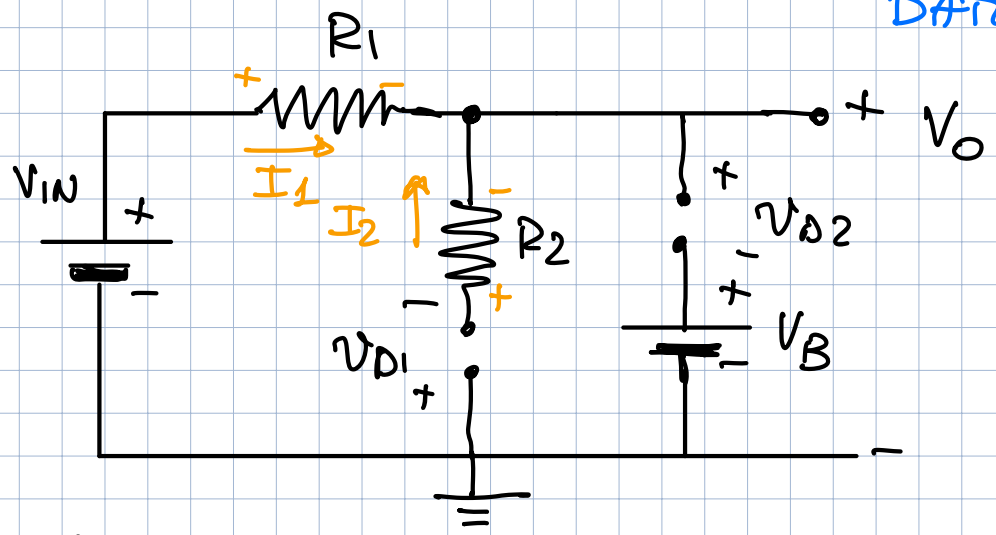
$v_{D1} = -V_o = -15V \Rightarrow$ OK $D1 = OFF$

QUINDI $V_{IN} = 30\,V$, $D1 = OFF$, $D2 = ON$ $\quad V_O = V_B = 15\,V$

FACOLTATIVO – TRANSCARATTERISTICA

HP $D1 = OFF$ e $D2 = OFF$

SEMPRE SFRUTTANDO I DATI PRECEDENTI



$I_1 = I_2 = 0$!

$V_O = V_{IN}$

$v_{D1} = -V_{IN}$

$v_{D2} = V_O - V_B \overset{!}{=} V_{IN} - V_B$

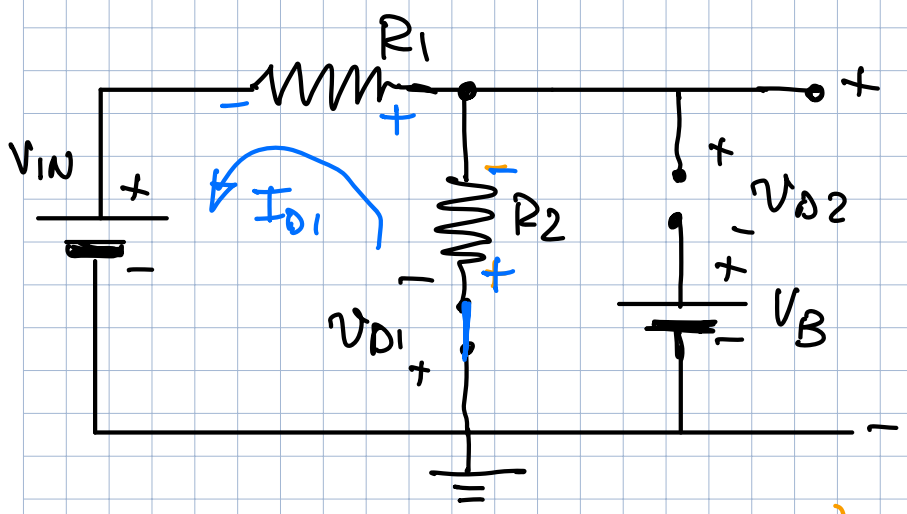
$v_{D1} < 0 \Rightarrow V_{IN} > 0$

$v_{D2} < 0 \Rightarrow V_{IN} - V_B < 0 \qquad V_{IN} < V_B$

$\Rightarrow \quad 0 \le V_{IN} \le V_B \qquad D1 = OFF,\ D2 = OFF \qquad V_O = V_{IN}$

RACCORDI: $V_O(0) = 0$; $V_O(V_B) = V_B = 15\,V$

CON $V_{IN} < 0 \Rightarrow$ D1 DIVENTA "ON"



$I_{D1}(R_1 + R_2) + V_{IN} = 0$

$I_{D1} = -\dfrac{V_{IN}}{R_1 + R_2} > 0$

$\Rightarrow V_{IN} < 0$

$V_O = -R_2 I_{D1} = +V_{IN}\dfrac{R_2}{R_1 + R_2}$

$\overset{!}{=} +V_{IN}/2$

RACCORDO: $V_O(0) = 0$ ✓

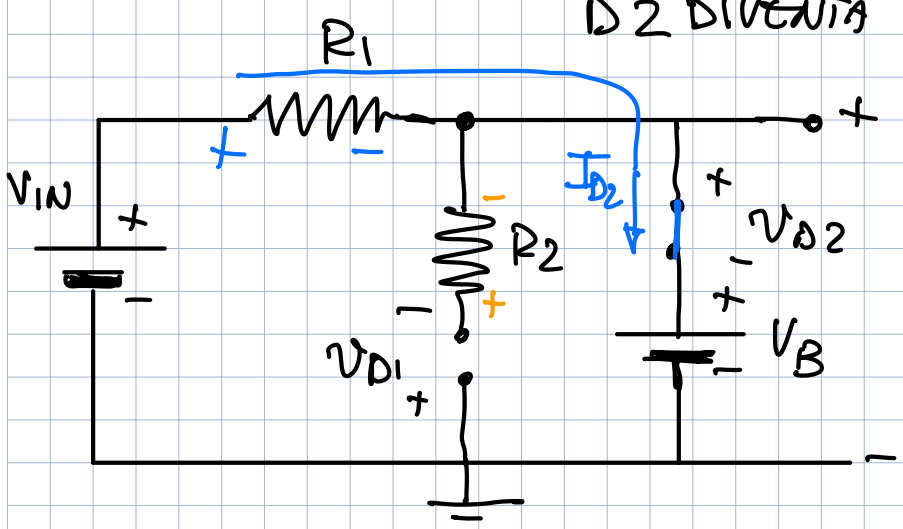
$\Rightarrow \quad v_{D2} = V_O - V_B < 0 \quad \Rightarrow \quad V_O < V_B$

$\Rightarrow \quad +\frac{V_{IN}}{2} < V_B \quad \Rightarrow \quad V_{IN} < 2V_B$ OK

$\Rightarrow$ $V_{IN} \leq 0$ $\quad$ D1 = ON, D2 = OFF $\quad$ $V_O = \frac{V_{IN}}{2}$

CON $V_{IN} > V_B$ $\Rightarrow$ RISPETTO AL CASO D1=D2=OFF

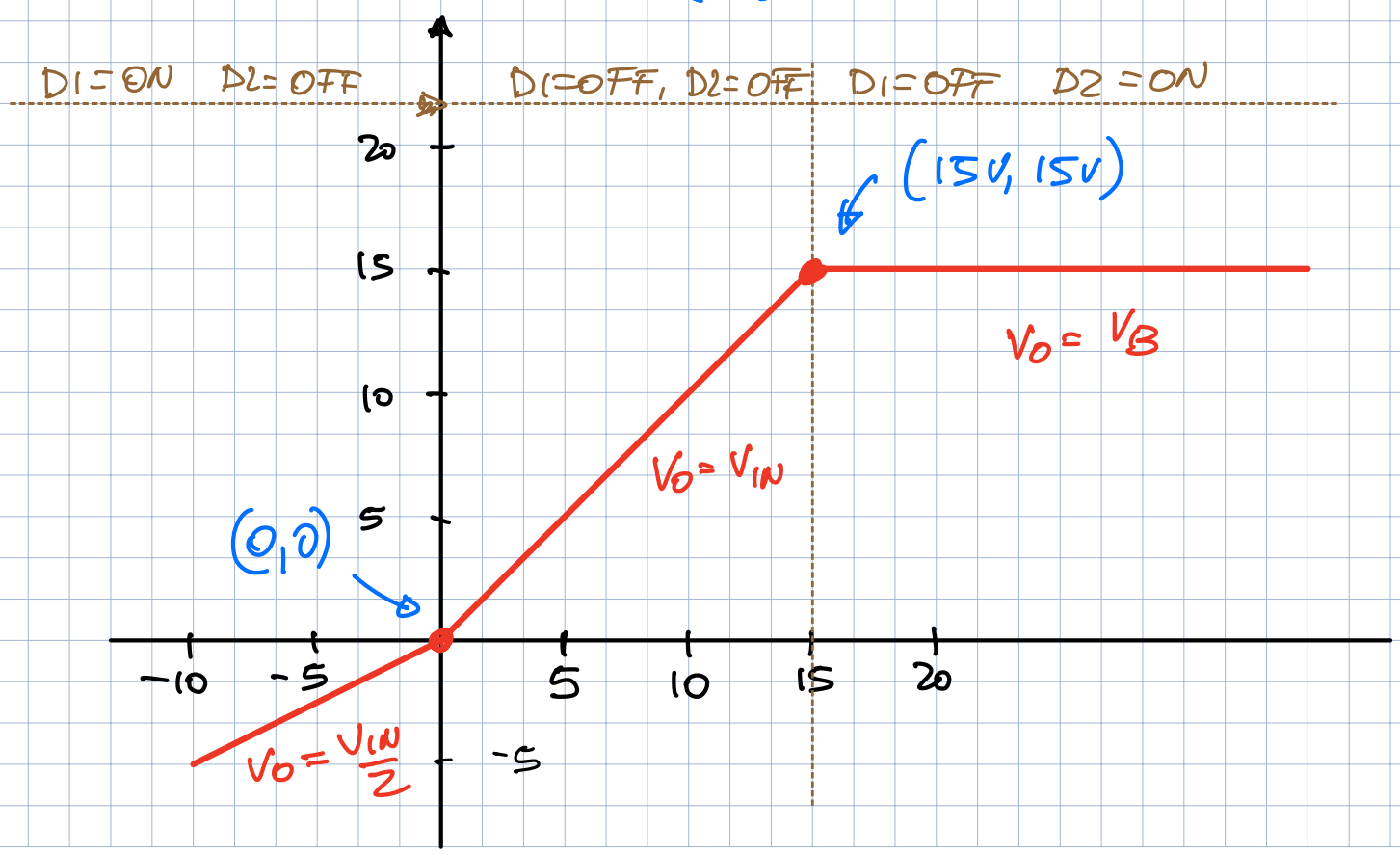
D2 DIVENTA "ON"



$I_{D2} = \frac{V_{IN} - V_B}{R_1} > 0$

PER $V_{IN} > V_B$ OK

$V_O = V_B = 15V$

$v_{D1} = -V_O = -15V \Rightarrow$ OK D1 = OFF

RACCORDO $V_O(15) = 15V$ ✓